

Final Project Proposal: Contextual Bandit Methods for Personalized Recommendation

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1. Introduction

Personalized recommendation systems are a core component of modern online platforms, aiming to adapt item suggestions to individual user preferences based on observed characteristics and historical interactions. More traditional approaches, such as collaborative filtering and matrix factorization, are typically optimized in offline settings and focus on static prediction accuracy. While effective in batch learning scenarios, these methods do not naturally account for sequential decision-making, uncertainty in user feedback, or the need for online adaptation.

Contextual multi-armed bandits provide a principled framework for modeling recommendation as an online learning problem. In this setting, an agent repeatedly observes a context vector describing the current user and sequentially selects an item to recommend, receiving feedback in the form of a reward. By explicitly balancing exploration and exploitation, contextual bandits naturally capture the inherent trade-offs in personalized recommendation systems and enable continual learning from interaction data. They were first successfully applied to personalized recommendation by Li et al. [1], who demonstrated their effectiveness in news article recommendation using logged interaction data. More recent work by Ban et. al [2] has extended this paradigm, namely in the form of neural contextual bandit models for personalized recommendation. These studies motivate a systematic comparison of standard contextual bandit algorithms in realistic recommendation settings. In this project, we will try to try to address:

How do different contextual bandit algorithms compare in terms of cumulative regret and convergence speed when learning personalized recommendations?

2. Motivation

There are two primary motivations for this work. First, contextual bandits provide a natural abstraction for recommendation systems in which user features (e.g., demographics, historical behavior, preferences) are available at decision time, but in which the utility of a recommendation is uncertain. By conditioning decisions on contextual information, these algorithms can generalize across users and adapt to evolving user preferences.

Second, while many contextual bandit algorithms come with strong theoretical guarantees, their empirical performance can vary depending on practical factors such as context dimensionality, feature representation, and evaluation metrics. If the observed context at time step t is represented by a feature vector $x_t \in \mathbb{R}^d$, the dimension d can affect estimation variance, sample complexity, and convergence speed. As a result, algorithms with similar regret bounds may behave differently when applied to real recommendation datasets.

To study these effects in a controlled yet realistic setting, this project will use the MovieLens 10M dataset [3], which contains over ten million movie ratings from thousands of users along with demographic information (it is publicly available at: <https://grouplens.org/datasets/movielens/10m/>). Logged user-item interactions will be treated as bandit feedback, enabling offline evaluation of learning algorithms without requiring live experimentation.

3. Methodology and Planned Contributions

This project formulates personalized recommendation as a contextual multi-armed bandit problem, and will proceed in the following steps:

- **Problem Formulation:** We will model our recommendation task as a contextual multi-armed bandit problem. At each interaction, the agent will observe a user context vector and select an item (arm) to recommend. The context vectors will be constructed from available demographic features and interaction statistics.
- **Algorithmic Techniques:** Several standard contextual bandit algorithms will be implemented and compared, including Linear Upper Confidence Bound (linear UCB), contextual Thompson sampling, and an ε -greedy baseline adapted to contextual settings. These methods represent different strategies for balancing exploration and exploitation, and are commonly referred to in the contextual-bandit literature.
- **Evaluation:** Since fully online experimentation is often impractical or infeasible, the project will rely on offline replay evaluation given logged recommendation data. At each time step, an algorithm selects an item given the observed context and receives a reward only if its chosen item matches the logged action. Model updates are performed only on matched events, ensuring fair and reproducible comparison across the various methods.
- **Analysis:** Our algorithms will be evaluated in terms of cumulative regret, learning curves and convergence behavior. Our analysis will see how well each method adapts to user preferences and how performance varies across users and items.
- **Insights:** We hope our final outcome to be an empirical study highlighting the trade-offs between exploration efficiency, stability, and scalability in the mentioned algorithms. We may look into extensions like richer feature representations or more thorough sensitivity analysis, where appropriate.

We hope this project yields an empirical comparison that highlights practical trade-offs between commonly used contextual bandit algorithms for personalized recommendation.

References

- [1] L. Li, W. Chu, J. Langford, and R. Schapire. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th International World Wide Web Conference (WWW)*, 2010. Available at: <http://dx.doi.org/10.1145/1772690.1772758>.
- [2] Yikun Ban, Yunzhe Qi, and Jingrui He. Neural contextual bandits for personalized recommendation. *arXiv preprint arXiv:2312.14037*, 2023. Available at: <https://arxiv.org/abs/2312.14037>.
- [3] GroupLens Research. MovieLens 10M Dataset. 2009. Available at: <https://grouplens.org/datasets/movielens/10m/>.